

A Separable-Range Duality for Strictly Cosingular Operators

Draft for verification – June 2026

Abstract

Let X and Y be Banach spaces and let $T: X \rightarrow Y$ be bounded. We prove that, when Y is separable, T is strictly cosingular if and only if its adjoint $T^*: Y^* \rightarrow X^*$ is strictly singular. In particular, separability of X is not needed. The main ingredient is a diagonal “preadjoint extraction” lemma: if $E \subseteq X^*$ is an infinite-dimensional closed subspace and $U: E \rightarrow Y^*$ is bounded, then, after passing to an infinite-dimensional closed subspace $E_0 \subseteq E$, the restriction $U|_{E_0}$ is induced by the adjoint of an operator from Y into the quotient $X/(E_0)^\perp$.

Status. This note records a proposed proof of a question that has appeared in the literature and on MathOverflow. The argument is written with all finite-dimensional interpolation, weak-star density, and quotient identifications made explicit so that it can be checked independently before being cited.

1 Introduction

Throughout, Banach spaces are over $\mathbb{K}$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and all subspaces occurring in definitions are understood to be closed.

An operator $S: E \rightarrow F$ is *strictly singular* if it is not bounded below on any infinite-dimensional subspace of E . Equivalently, no restriction of S to an infinite-dimensional subspace is an isomorphism onto its image.

An operator $T: X \rightarrow Y$ is *strictly cosingular* if, for every infinite-codimensional subspace $M \subseteq Y$, the operator

$$Q_M T: X \rightarrow Y/M$$

is not surjective, where $Q_M: Y \rightarrow Y/M$ is the quotient map. Pełczyński introduced and studied these classes in [3]. It is standard that

$$T^* \text{ strictly singular} \implies T \text{ strictly cosingular}.$$

The converse is false without additional assumptions: the canonical inclusion $c_0 \hookrightarrow \ell_\infty$ is strictly cosingular, while its adjoint is not strictly singular. The question considered here asks whether the converse holds when the range space Y is separable; see [1, 2].

Our main result is the following.

Theorem 1.1. *Let X be a Banach space, let Y be a separable Banach space, and let $T: X \rightarrow Y$ be bounded. Then*

$$T \text{ is strictly cosingular} \iff T^* \text{ is strictly singular}.$$

The proof does not seek a well-behaved basic sequence inside a subspace on which T^* is bounded below. Instead, it passes to the image of that subspace under T^* and uses separability of Y to force the inverse map, on a further infinite-dimensional subspace, to be the restriction of an adjoint. The resulting weak-star closed witness need not be contained in the original subspace.

2 Preliminaries

For a subset $E \subseteq X^*$ and a subset $M \subseteq X$, write

$$E_\perp = \{x \in X : e(x) = 0 \text{ for every } e \in E\},$$

$$M^\perp = \{x^* \in X^* : x^*(m) = 0 \text{ for every } m \in M\}.$$

We use $\sigma(X^*, X)$ for the weak-star topology on X^* .

Lemma 2.1 (Surjectivity and the adjoint). *Let $S: X \rightarrow Z$ be bounded. The following are equivalent:*

- (i) S is surjective;
- (ii) S^* is bounded below, that is, there is $c > 0$ such that

$$\|S^*z^*\| \geq c\|z^*\| \quad (z^* \in Z^*).$$

Proof. If S is surjective, the open mapping theorem gives $C > 0$ such that every $z \in Z$ has a preimage $x \in X$ with $\|x\| \leq C\|z\|$. Hence, for $z^* \in Z^*$,

$$\|z^*\| = \sup_{\|z\| \leq 1} |z^*(z)| \leq C \sup_{\|x\| \leq 1} |z^*(Sx)| = C\|S^*z^*\|.$$

Conversely, if S^* is bounded below, then S^* is injective and has closed range. The closed range theorem implies that $\text{ran } S$ is closed, while

$$\overline{\text{ran } S} = Z$$

because $(\text{ran } S)^\perp = \ker S^* = \{0\}$. Thus S is surjective. □

Proposition 2.2 (Weak-star characterization of strict cosingularity). *For a bounded operator $T: X \rightarrow Y$, the following are equivalent:*

- (i) T is strictly cosingular;
- (ii) T^* is not bounded below on any infinite-dimensional weak-star closed subspace of Y^* .

Proof. If $M \subseteq Y$ is closed, then Q_M^* identifies $(Y/M)^*$ isometrically and weak-star homeomorphically with $M^\perp \subseteq Y^*$. Moreover,

$$(Q_M T)^* = T^* Q_M^*.$$

By Lemma 2.1, $Q_M T$ is surjective if and only if T^* is bounded below on $M^\perp$. Finally, every weak-star closed subspace $W \subseteq Y^*$ has the form

$$W = (W_\perp)^\perp,$$

and W is infinite-dimensional if and only if $Y/W_\perp$ is infinite-dimensional. This gives the equivalence. □

We next isolate the finite-dimensional interpolation fact used in the diagonal construction.

Lemma 2.3 (Finite-dimensional interpolation). *Let $F \subseteq X^*$ be finite-dimensional and let $\varphi \in F^*$. For every $\eta > 0$ there exists $x \in X$ such that*

$$f(x) = \varphi(f) \quad (f \in F)$$

and

$$\|x\| \leq (1 + \eta)\|\varphi\|.$$

For $\varphi = 0$, one may take $x = 0$.

Proof. Define

$$R_F: X \longrightarrow F^*, \quad (R_F x)(f) = f(x).$$

The adjoint $R_F^*: F^{**} \rightarrow X^*$ is, under the canonical identification $F^{**} = F$, the inclusion $F \hookrightarrow X^*$; in particular, it is an isometry. It follows first that R_F is onto. Indeed, if its range were a proper subspace of the finite-dimensional space F^* , some nonzero element of $F = (F^*)^*$ would annihilate $\text{ran } R_F$, contradicting injectivity of R_F^* .

Equip F^* with the quotient norm

$$\|\psi\|_q = \inf\{\|x\| : R_F x = \psi\}.$$

The dual norm induced on F is

$$\|f\|_{q,*} = \|R_F^* f\| = \|f\|.$$

The original norm on F^* has the same dual norm on F . Since a norm is recovered from its dual unit ball, the quotient norm and the original norm on F^* coincide. Thus

$$\inf\{\|x\| : R_F x = \varphi\} = \|\varphi\|,$$

and the asserted choice of x follows. □

3 The preadjoint extraction lemma

The following is the central device.

Lemma 3.1 (Preadjoint extraction). *Let Y be separable, let $E \subseteq X^*$ be a closed infinite-dimensional subspace, and let*

$$U: E \longrightarrow Y^*$$

be bounded. Then there exist

- *a closed infinite-dimensional subspace $E_0 \subseteq E$;*
- *the quotient space*

$$P = X/(E_0)_\perp$$

with quotient map $q: X \rightarrow P$;

- *an operator $A: Y \rightarrow P$ with $\|A\| \leq 2\|U\|$;*

such that the following holds. For $e \in E_0$, let $\hat{e} \in P^*$ be the functional

$$\hat{e}(qx) = e(x) \quad (x \in X).$$

Then

$$A^*\hat{e} = Ue \quad (e \in E_0). \quad (1)$$

Proof. Let

$$\mathbb{F} = \begin{cases} \mathbb{Q}, & \mathbb{K} = \mathbb{R}, \\ \mathbb{Q} + i\mathbb{Q}, & \mathbb{K} = \mathbb{C}. \end{cases}$$

Choose a countable dense $\mathbb{F}$ -linear subspace $D \subseteq Y$, and enumerate all of its elements as

$$D = \{y_1, y_2, \dots\}.$$

We recursively construct linearly independent unit vectors $e_n \in E$ and vectors $x_n \in X$.

Suppose that $e_1, \dots, e_{n-1}$ have been chosen, and put

$$F_{n-1} = \text{span}\{e_1, \dots, e_{n-1}\}.$$

Define $\varphi_n \in F_{n-1}^*$ by

$$\varphi_n(e) = (Ue)(y_n), \quad e \in F_{n-1}.$$

Then

$$\|\varphi_n\| \leq \|U\| \|y_n\|.$$

By Lemma 2.3, choose $x_n \in X$ such that

$$e(x_n) = (Ue)(y_n) \quad (e \in F_{n-1}) \quad (2)$$

and

$$\|x_n\| \leq 2\|U\| \|y_n\|. \quad (3)$$

When $y_n = 0$, take $x_n = 0$.

Define $h_n \in E^*$ by

$$h_n(e) = (Ue)(y_n) - e(x_n), \quad e \in E.$$

The subspace

$$H_n = \bigcap_{j=1}^n \ker h_j$$

has finite codimension in the infinite-dimensional space E , and is therefore infinite-dimensional. Choose

$$e_n \in H_n \setminus F_{n-1}$$

with $\|e_n\| = 1$. This completes the recursion.

Set

$$E_0 = \overline{\text{span}}\{e_n : n \geq 1\} \subseteq E.$$

The sequence (e_n) is linearly independent, so E_0 is infinite-dimensional. We claim that

$$(Ue)(y_j) = e(x_j) \quad (e \in E_0, j \geq 1). \quad (4)$$

It suffices first to check this for $e = e_i$. If $i < j$, then $e_i \in F_{j-1}$ and (2) gives $h_j(e_i) = 0$. If $i \geq j$, then $e_i \in H_i \subseteq \ker h_j$, again giving $h_j(e_i) = 0$. Thus h_j vanishes on every e_i , and by continuity it vanishes on E_0 . This proves (4).

Let

$$P = X/(E_0)_\perp$$

and let $q: X \rightarrow P$ be the quotient map. Define a map $A_0: D \rightarrow P$ by prescribing

$$A_0 y_n = q x_n \quad (n \geq 1). \quad (5)$$

We verify that this prescription is well-defined and $\mathbb{F}$ -linear. Suppose

$$\sum_{j=1}^m a_j y_{n_j} = 0, \quad a_j \in \mathbb{F}.$$

For every $e \in E_0$, equation (4) gives

$$e \left(\sum_{j=1}^m a_j x_{n_j} \right) = \sum_{j=1}^m a_j (Ue)(y_{n_j}) = (Ue) \left(\sum_{j=1}^m a_j y_{n_j} \right) = 0.$$

Therefore $\sum_{j=1}^m a_j x_{n_j} \in (E_0)_\perp$, and hence

$$\sum_{j=1}^m a_j q x_{n_j} = 0.$$

Thus all $\mathbb{F}$ -linear relations in D are respected, proving that A_0 is well-defined and $\mathbb{F}$ -linear.

For every $y \in D$, choose n with $y = y_n$. By (3),

$$\|A_0 y\| = \|q x_n\| \leq \|x_n\| \leq 2\|U\| \|y\|.$$

Consequently, A_0 extends uniquely to a bounded additive map

$$A: Y \longrightarrow P$$

with $\|A\| \leq 2\|U\|$. The extension is $\mathbb{K}$ -linear: in the real case this follows by approximating real scalars by rationals, and in the complex case by approximating complex scalars by elements of $\mathbb{Q} + i\mathbb{Q}$.

Finally, fix $e \in E_0$. Since e vanishes on $(E_0)_\perp$, it induces the functional $\hat{e} \in P^*$ appearing in the statement. For every n ,

$$(A^* \hat{e})(y_n) = \hat{e}(A y_n) = \hat{e}(q x_n) = e(x_n) = (Ue)(y_n).$$

The set D is dense in Y , so the two continuous functionals $A^* \hat{e}$ and Ue agree on all of Y . This proves (1). $\square$

Remark 3.2. No basis property is required of the sequence (e_n) . Linear independence is enough, since only the infinite dimensionality of $E_0 = \overline{\text{span}}\{e_n\}$ is used. Thus the argument does not split according to whether the original subspace contains an unconditional or a hereditarily indecomposable subspace.

4 Proof of the main theorem

We first prove the nontrivial implication in a factorized form.

Theorem 4.1 (Quotient factorization). *Let Y be separable and let $T: X \rightarrow Y$ be bounded. If T^* is bounded below on some infinite-dimensional subspace of Y^* , then there exist an infinite-dimensional Banach space P , a quotient map $q: X \rightarrow P$, and an operator $A: Y \rightarrow P$ such that*

$$AT = q. \quad (6)$$

In particular, T is not strictly cosingular.

Proof. Suppose that T^* is bounded below on an infinite-dimensional subspace of Y^* . Passing to its norm closure, choose a closed infinite-dimensional subspace $Z \subseteq Y^*$ and $c > 0$ such that

$$\|T^*z\| \geq c\|z\| \quad (z \in Z).$$

Then

$$E = T^*(Z) \subseteq X^*$$

is closed and infinite-dimensional, and

$$U = (T^*|_Z)^{-1}: E \longrightarrow Z \subseteq Y^*$$

is bounded.

Apply Lemma 3.1 to E and U . We obtain a closed infinite-dimensional subspace $E_0 \subseteq E$, the quotient

$$P = X/(E_0)_\perp,$$

the quotient map $q: X \rightarrow P$, and an operator $A: Y \rightarrow P$ such that

$$A^*\hat{e} = Ue \quad (e \in E_0).$$

For $e \in E_0$,

$$(AT)^*\hat{e} = T^*A^*\hat{e} = T^*Ue = e = q^*\hat{e}. \quad (7)$$

It remains to extend (7) from E_0 to all of P^* . The adjoint q^* is an isometric weak-star homeomorphism from P^* onto

$$q^*(P^*) = ((E_0)_\perp)^\perp.$$

By the bipolar theorem for annihilators,

$$((E_0)_\perp)^\perp = \overline{E_0}^{\sigma(X^*, X)}. \quad (8)$$

Hence

$$\hat{E}_0 := \{\hat{e} : e \in E_0\}$$

is weak-star dense in P^* . Both

$$(AT)^*, q^*: P^* \longrightarrow X^*$$

are weak-star-to-weak-star continuous. Therefore (7) extends to every $p^* \in P^*$:

$$(AT)^*p^* = q^*p^*.$$

Dual functionals separate points of P , so this equality of adjoints implies

$$AT = q.$$

Since q is surjective, (6) implies that A is surjective. Put $M = \ker A$. The operator A induces an isomorphism

$$\tilde{A}: Y/M \longrightarrow P.$$

The space P is infinite-dimensional because P^* contains the infinite-dimensional subspace $\hat{E}_0$. Thus M has infinite codimension in Y . Moreover,

$$\tilde{A} Q_M T = AT = q,$$

so

$$Q_M T = \tilde{A}^{-1} q$$

is surjective. Hence T is not strictly cosingular. $\square$

Proof of Theorem 1.1. If T^* is strictly singular, then it is not bounded below on any infinite-dimensional weak-star closed subspace of Y^* , so Proposition 2.2 shows that T is strictly cosingular.

Conversely, assume that T is strictly cosingular. If T^* were not strictly singular, it would be bounded below on some infinite-dimensional subspace of Y^* . Theorem 4.1 would then imply that T is not strictly cosingular, a contradiction. Therefore T^* is strictly singular. $\square$

The argument also gives the geometric formulation sought in the original question.

Corollary 4.2 (Weak-star extraction). *Let Y be separable and let $T: X \rightarrow Y$ be bounded. If T^* is bounded below on an infinite-dimensional norm-closed subspace of Y^* , then T^* is bounded below on an infinite-dimensional weak-star closed subspace of Y^* .*

Proof. Use the notation supplied by Theorem 4.1. Since $A: Y \rightarrow P$ is surjective,

$$W = A^*(P^*) = (\ker A)^\perp$$

is an infinite-dimensional weak-star closed subspace of Y^* . The identity $AT = q$ gives

$$T^* A^* = q^*.$$

For $p^* \in P^*$,

$$\|T^* A^* p^*\| = \|q^* p^*\| = \|p^*\| \geq \frac{1}{\|A\|} \|A^* p^*\|.$$

Thus T^* is bounded below on W . $\square$

5 Remarks on the proof

Remark 5.1 (Where separability is used). Separability of Y is used only in Lemma 3.1, through the existence of the countable dense $\mathbb{F}$ -linear subspace $D \subseteq Y$. No separability assumption on X occurs anywhere in the proof.

Remark 5.2 (Why the classical counterexample is not covered). The inclusion $c_0 \hookrightarrow \ell_\infty$ has nonseparable range space ℓ_∞ , so the countable diagonal construction in Lemma 3.1 is unavailable. Thus the theorem is consistent with Pełczyński's counterexample.

Remark 5.3 (The new witness need not lie in the old one). Starting from a subspace $Z \subseteq Y^*$ on which T^* is bounded below, the proof does not attempt to find a weak-star closed subspace inside Z . It first passes to $E = T^*(Z)$, extracts $E_0 \subseteq E$, and then constructs

$$W = A^*(P^*) \subseteq Y^*.$$

The space W is weak-star closed and is preserved by T^* , but need not be contained in Z or contain Z . This is the point that avoids structural assumptions on subspaces of Y^* .

6 Proof audit

For ease of independent checking, the logical dependencies of the proof are as follows.

1. Lemma 2.3 uses only finite-dimensional duality. In particular, the representing vector $x_n \in X$ is exact on the finite-dimensional space F_{n-1} , while its norm is within an arbitrary factor $1 + \eta$ of the optimal norm.
2. At the n th recursive step of Lemma 3.1, only finitely many kernels $\ker h_1, \dots, \ker h_n$ are imposed. Their intersection is finite-codimensional and therefore remains infinite-dimensional; this permits the choice $e_n \notin \text{span}\{e_1, \dots, e_{n-1}\}$.
3. Equation (4) is checked separately for $i < j$ and $i \geq j$. This is the sole diagonalization step.
4. The enumeration of the entire $\mathbb{F}$ -linear space D , rather than merely a dense sequence, is what yields the uniform estimate

$$\|A_0 y\| \leq 2\|U\|\|y\| \quad (y \in D)$$

after linear relations have been verified.

5. The passage from equality on $\widehat{E}_0$ to equality on P^* uses precisely

$$\overline{E_0}^{w*} = ((E_0)_\perp)^\perp = q^*(P^*)$$

and weak-star continuity of adjoints. No norm-density assertion is made or needed.

6. Once $AT = q$, surjectivity of q forces A to be surjective, and the quotient $Y/\ker A$ is isomorphic to the infinite-dimensional space P . This produces the explicit infinite-codimensional quotient on which T is onto.

References

- [1] G. Androulakis and K. Beanland, *Descriptive set theoretic methods applied to strictly singular and strictly cosingular operators*, Quaestiones Math. **31** (2008), 151–161.

- [2] K. Beanland, *Strictly singular operators and their adjoints*, MathOverflow, Question 98449 (2012), [online discussion](#).
- [3] A. Pełczyński, *On strictly singular and strictly cosingular operators. I. Strictly singular and strictly cosingular operators in $C(S)$ -spaces*, Bull. Acad. Polon. Sci. Sér. Sci. Math. Astronom. Phys. **13** (1965), 31–36.